Active Inference for Joint Port Selection and Pilot Allocation in Fluid Antenna Systems

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***Abstract*—*[DRAFT — rewrite last, once the results section is frozen.]* Fluid antenna systems (FAS) obtain spatial degrees of freedom by activating a small subset of a dense grid of candidate ports, but selecting that subset presupposes channel knowledge across ports that are never observed. Existing port-selection methods assume full or near-full channel state information (CSI) and treat pilot overhead as exogenous. We instead formulate port selection and pilot allocation jointly as *active inference*: a spatio-temporal generative model maintains a Gaussian belief over all N port channels, and the transmitter selects both the ports it serves and the ports it pilots by minimizing an expected free energy (EFE) that trades achievable rate against information gain and port-switching cost. [TODO: one sentence on the temporal model / online Doppler learning] On a 21×21 port grid, the proposed scheme attains a higher sum rate while piloting only 60% of the active ports than a memoryless baseline piloting all of them, and recovers 83% of the oracle rate when the Doppler rate is unknown and learned online.**

***Index Terms*—Fluid antenna systems, port selection, active inference, expected free energy, pilot allocation, partial CSI.**

I. Introduction

[TODO: Para 1: FAS context — ports, spatial DoF, RF-chain budget.]

[TODO: Para 2: the acquisition problem — scoring N ports needs CSI at N ports; overhead grows with the grid.]

[TODO: Para 3: prior work and the gap — learning-based selection (DRL, bandits) assumes full CSI; FAS channel estimation solves *estimation*, not the *decision*. No prior work allocates pilots as part of the selection decision.]

[TODO: Para 4: contributions, as a bulleted list.]

II. System Model

A. Port Geometry and Spatial Correlation

The base station is equipped with a fluid antenna whose radiating element may be positioned at any of N candidate *ports* arranged on a uniform $N_x \times N_y$ grid over an aperture of $W_x\lambda \times W_y\lambda$, with λ the wavelength. Let $\mathbf{u}_n \in \mathbb{R}^2$ denote the position of port n and $d_{ij} = \|\mathbf{u}_i - \mathbf{u}_j\|$. Under the classical Jakes model for two-dimensional isotropic scattering, the spatial correlation between two ports depends only on their separation,

$$[\mathbf{R}]_{ij} \;=\; J_0\!\left(\frac{2\pi d_{ij}}{\lambda}\right), \tag{1}$$

where $J_0(\cdot)$ is the zeroth-order Bessel function of the first kind. The channel from the N ports to user k is spatially correlated Rayleigh, $\mathbf{h}_k(t) \in \mathbb{C}^N$ with $\mathbf{h}_k(t) \sim \mathcal{CN}(\mathbf{0}, \beta_k\mathbf{R})$, where β_k is the large-scale gain of user k. Because the port spacing is sub-wavelength, $\mathbf{R}$ is strongly rank-deficient; Section III exploits this.

B. Temporal Model

The channel evolves across slots as an order-p autoregressive process driven by spatially correlated innovations,

$$\mathbf{h}_k(t) = \sum_{i=1}^{p} a_i\,\mathbf{h}_k(t-i) + \mathbf{e}_k(t), \quad \mathbf{e}_k(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_v^2\beta_k\mathbf{R}), \tag{2}$$

with $\mathbf{e}_k(t)$ independent across t and of the past. The coefficients $\mathbf{a} = [a_1, \dots, a_p]^{\mathrm{T}}$ and the innovation variance σ_v^2 are those of the Jakes temporal autocorrelation $r(\tau) = J_0(2\pi f_D T_s \tau)$, obtained from the Yule–Walker system

$$\boldsymbol{\Gamma}\mathbf{a} = \mathbf{r}, \qquad \sigma_v^2 = r(0) - \mathbf{a}^{\mathrm{T}}\mathbf{r}, \tag{3}$$

where $[\boldsymbol{\Gamma}]_{ij} = r(|i-j|)$ and $\mathbf{r} = [r(1), \dots, r(p)]^{\mathrm{T}}$. Here f_D is the maximum Doppler frequency and T_s the slot duration. Setting $p = 1$ recovers the first-order Gauss–Markov model $a_1 = \rho = J_0(2\pi f_D T_s)$ used in most prior work; the value of $p > 1$ is quantified in Section VI.

C. Activation and Pilot Allocation

In slot t the agent activates a set $\mathcal{S}_t \subset \{1, \dots, N\}$ of $|\mathcal{S}_t| = M$ ports, and *pilots a subset of them*,

$$\mathcal{Q}_t \subseteq \mathcal{S}_t, \qquad |\mathcal{Q}_t| = m \le M \le N. \tag{4}$$

For an index set $\mathcal{A}$ let $\mathbf{P}_{\mathcal{A}} \in \{0,1\}^{|\mathcal{A}|\times N}$ be the selection matrix whose rows are the standard basis vectors indexed by $\mathcal{A}$, so $\mathbf{P}_{\mathcal{A}}\mathbf{x}$ extracts the entries of $\mathbf{x}$ on $\mathcal{A}$. During the pilot phase the users transmit orthogonal sequences and the base station observes

$$\mathbf{y}_k(t) \;=\; \mathbf{P}_{\mathcal{Q}_t}\mathbf{h}_k(t) + \mathbf{n}_k(t), \qquad \mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_e^2\mathbf{I}_m), \tag{5}$$

with σ_e^2 set by the pilot SNR. The remaining $N - m$ ports — including the $M - m$ ports that are activated but *not* piloted — are never measured in slot t and must be inferred from (1) and (2). Prior port-selection work fixes $m = M$; treating $m < M$ makes the pilot budget a design variable, which is the object of this letter.

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D. Hybrid Downlink Transmission

The M activated ports are driven by $n_{\mathrm{RF}} \leq M$ radio-frequency chains through a fully-connected network of unit-modulus phase shifters. The transmitted signal is $\mathbf{x}(t) = \mathbf{F}_{\mathrm{RF}}\mathbf{W}_{\mathrm{BB}}\mathbf{s}(t)$ with $\mathbf{s} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_K)$, giving the effective precoder

$$\mathbf{W} = \mathbf{F}_{\mathrm{RF}}\mathbf{W}_{\mathrm{BB}} \in \mathbb{C}^{M\times K}, \quad \big|[\mathbf{F}_{\mathrm{RF}}]_{ij}\big| = 1, \quad \|\mathbf{W}\|_F^2 \leq P. \tag{6}$$

Writing $\mathbf{h}_{k,\mathcal{S}} = \mathbf{P}_{\mathcal{S}}\mathbf{h}_k$ and treating inter-user interference as noise, user k achieves

$$R_k \;=\; \log_2\!\left(1 + \frac{\big|\mathbf{h}_{k,\mathcal{S}}^{\mathrm{H}}\mathbf{w}_k\big|^2}{\sum_{j\neq k}\big|\mathbf{h}_{k,\mathcal{S}}^{\mathrm{H}}\mathbf{w}_j\big|^2 + \sigma^2}\right), \tag{7}$$

where σ^2 is the receiver noise power and $\mathbf{w}_k$ the k-th column of $\mathbf{W}$.

E. Problem Statement

Repositioning the element between slots costs delay and energy, which we charge through the number of ports that change, $C_t = |\mathcal{S}_t \triangle \mathcal{S}_{t-1}|$, with $\triangle$ the symmetric set difference. The design problem is

$$\begin{aligned} \max_{\{\mathcal{S}_t,\,\mathcal{Q}_t\}} \;& \sum_{t=1}^{T}\left(\sum_{k=1}^{K} R_k(t) - \eta_{\mathrm{sw}} C_t\right) \\ \text{s.t. } \; & |\mathcal{S}_t| = M, \quad \mathcal{Q}_t \subseteq \mathcal{S}_t, \quad |\mathcal{Q}_t| = m, \end{aligned} \tag{8}$$

with $\eta_{\mathrm{sw}} \geq 0$ trading throughput against reconfiguration. The difficulty is that $R_k(t)$ depends on the channel at ports that (5) never reveals, so $\mathcal{S}_t$ and $\mathcal{Q}_t$ must be chosen from a *belief* rather than from the channel.

III. Generative Model and Belief

A. State-Space Form

Stacking the last p channel snapshots into $\mathbf{z}_k(t) = [\mathbf{h}_k(t)^{\mathrm{T}}, \ldots, \mathbf{h}_k(t-p+1)^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{C}^{pN}$ turns (2) into a first-order recursion

$$\mathbf{z}_k(t) = \mathbf{F}\,\mathbf{z}_k(t-1) + \boldsymbol{\epsilon}_k(t), \qquad \mathbf{F} = \mathbf{C}_{\mathbf{a}} \otimes \mathbf{I}_N, \tag{9}$$

where $\mathbf{C}_{\mathbf{a}} \in \mathbb{R}^{p\times p}$ is the companion matrix of $\mathbf{a}$ and the process noise is confined to the current-time block,

$$\mathbf{Q}_k = \mathbb{E}\big[\boldsymbol{\epsilon}_k\boldsymbol{\epsilon}_k^{\mathrm{H}}\big] = \big(\mathbf{e}_1\mathbf{e}_1^{\mathrm{T}}\big) \otimes \big(\sigma_v^2\beta_k\mathbf{R}\big), \tag{10}$$

with $\mathbf{e}_1$ the first column of $\mathbf{I}_p$. The Kronecker structure states the model's key property: the temporal dynamics act identically on every port, and the spatial correlation is identical in every slot. The observation (5) reads $\mathbf{y}_k(t) = \mathbf{G}_{\mathcal{Q}_t}\mathbf{z}_k(t) + \mathbf{n}_k(t)$ with $\mathbf{G}_{\mathcal{A}} = [\,\mathbf{P}_{\mathcal{A}} \quad \mathbf{0}_{|\mathcal{A}|\times(p-1)N}\,]$, i.e. pilots see the current channel only.

B. Kalman Recursion

The model is linear-Gaussian, so the exact posterior is Gaussian, $q(\mathbf{z}_k) = \mathcal{CN}(\hat{\mathbf{z}}_k, \mathbf{P}_k)$, propagated by a per-user complex Kalman filter. The predict step advances the belief through the AR dynamics, encoding channel aging:

$$\hat{\mathbf{z}}_k \leftarrow \mathbf{F}\hat{\mathbf{z}}_k, \qquad \mathbf{P}_k \leftarrow \mathbf{F}\mathbf{P}_k\mathbf{F}^{\mathrm{H}} + \mathbf{Q}_k. \tag{11}$$

Given the pilots on $\mathcal{Q}_t$, with $\mathbf{G} \triangleq \mathbf{G}_{\mathcal{Q}_t}$ and Kalman gain

$$\mathbf{K}_k = \mathbf{P}_k\mathbf{G}^{\mathrm{H}}\big(\mathbf{G}\mathbf{P}_k\mathbf{G}^{\mathrm{H}} + \sigma_e^2\mathbf{I}_m\big)^{-1}, \tag{12}$$

the correction step is

$$\hat{\mathbf{z}}_k \leftarrow \hat{\mathbf{z}}_k + \mathbf{K}_k\big(\mathbf{y}_k - \mathbf{G}\hat{\mathbf{z}}_k\big), \tag{13}$$

$$\mathbf{P}_k \leftarrow \big(\mathbf{I} - \mathbf{K}_k\mathbf{G}\big)\mathbf{P}_k\big(\mathbf{I} - \mathbf{K}_k\mathbf{G}\big)^{\mathrm{H}} + \sigma_e^2\mathbf{K}_k\mathbf{K}_k^{\mathrm{H}}. \tag{14}$$

Only the $m \times m$ innovation covariance is inverted, and it is regularized by $\sigma_e^2\mathbf{I}_m$; the rank-deficient prior $\mathbf{R}$ therefore needs no artificial jitter, and the Joseph form (14) keeps $\mathbf{P}_k$ Hermitian positive semidefinite. The belief over the current channel is the leading block,

$$\boldsymbol{\mu}_k = \boldsymbol{\Pi}\hat{\mathbf{z}}_k, \qquad \boldsymbol{\Sigma}_k = \boldsymbol{\Pi}\mathbf{P}_k\boldsymbol{\Pi}^{\mathrm{H}}, \qquad \boldsymbol{\Pi} = [\,\mathbf{I}_N \;\; \mathbf{0}\,]. \tag{15}$$

Two couplings do the work. Spatially, the off-diagonals of $\mathbf{R}$ propagate a measurement at one port to its correlated neighbours, so m pilots inform far more than m ports. Temporally, (11) carries information forward across slots, so a port piloted in an earlier slot retains a usable estimate. Together they are what permit $m < M$.

C. Exact Reduced-Rank Implementation

Carrying $\mathbf{P}_k \in \mathbb{C}^{pN\times pN}$ is infeasible at the scale we target ($N = 441$, $p = 4$ gives 1764×1764 per user per slot). It is also unnecessary. Let $\mathbf{R} = \mathbf{B}\boldsymbol{\Lambda}\mathbf{B}^{\mathrm{H}}$ be the eigendecomposition truncated to the r eigenvalues carrying all but a fraction 10^{-6} of the energy. Since $\mathbf{h}_k(t) \in \mathrm{range}(\mathbf{R})$ for every t,

$$\mathbf{h}_k(t) = \mathbf{B}\,\mathbf{c}_k(t), \qquad \mathbf{c}_k(t) \in \mathbb{C}^r, \tag{16}$$

holds *exactly*, and $\mathbf{c}_k$ is a sufficient statistic for $\mathbf{h}_k$. Because (9) acts identically on every port, the dynamics carry over unchanged with $\mathbf{I}_N$ replaced by $\mathbf{I}_r$. The state dimension drops from pN to pr and the per-slot cost from $\mathcal{O}\big((pN)^3\big)$ to $\mathcal{O}\big((pr)^3\big)$. At $N = 441$ the numerical rank is $r = 26$, a reduction of roughly 4.9×10^3 in arithmetic at no modelling cost.

IV. Expected Free Energy and Selection

A. The Objective

Rather than treating exploration as a heuristic bolted onto a rate maximizer, we score a candidate pair $(\mathcal{S}, \mathcal{Q})$ by an expected free energy, written as a value to be minimized,

$$G(\mathcal{S}, \mathcal{Q}) = -\,\alpha\,\mathrm{Prag}(\mathcal{S}) - \,\beta_w\,\mathrm{Epis}(\mathcal{Q}) + \,\eta_{\mathrm{sw}}\big|\mathcal{S} \triangle \mathcal{S}_{t-1}\big|. \tag{17}$$

The *pragmatic* term is the sum rate the belief predicts, penalized by its own residual uncertainty. With $\boldsymbol{\mu}_{k,\mathcal{S}} = \mathbf{P}_{\mathcal{S}}\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_{k,\mathcal{S}} = \mathbf{P}_{\mathcal{S}}\boldsymbol{\Sigma}_k\mathbf{P}_{\mathcal{S}}^{\mathrm{H}}$,

$$\mathrm{Prag}(\mathcal{S}) = \sum_{k=1}^{K} \log_2\left(1 + \frac{\left|\boldsymbol{\mu}_{k,\mathcal{S}}^{\mathrm{H}}\mathbf{w}_k\right|^2}{\mathcal{I}_k}\right), \tag{18}$$

$$\mathcal{I}_k = \sum_{j\neq k}\left|\boldsymbol{\mu}_{k,\mathcal{S}}^{\mathrm{H}}\mathbf{w}_j\right|^2 + \underbrace{\sum_j \mathbf{w}_j^{\mathrm{H}}\boldsymbol{\Sigma}_{k,\mathcal{S}}\mathbf{w}_j}_{\text{CSI-error leakage}} + \sigma^2. \tag{19}$$

The middle term of (19) is what makes the agent conservative when uncertain: a port with a large posterior variance contributes interference it cannot precode away. It is also what the robust MMSE precoder used for $\mathbf{W}$ minimizes, before the hybrid factorization (6) is applied.

The *epistemic* term is the mutual information between the channel and the pilots that $\mathcal{Q}$ would produce, i.e. the expected reduction in belief entropy,

$$\mathrm{Epis}(\mathcal{Q}) = \sum_{k=1}^{K} \log_2\det\left(\mathbf{I}_m + \sigma_e^{-2}\,\mathbf{P}_{\mathcal{Q}}\boldsymbol{\Sigma}_k\mathbf{P}_{\mathcal{Q}}^{\mathrm{H}}\right). \tag{20}$$

Through the off-diagonals of $\boldsymbol{\Sigma}_k$, piloting one port also gathers information about its neighbours. The two terms encode the exploit–explore tension natively: (18) favours ports that serve well now, (20) favours ports that most reduce uncertainty for later, and β_w sets the balance.

B. Greedy Selection

Exact minimization of (17) is combinatorial. We build the sets greedily, adding at each step the port whose inclusion most decreases G,

$$p^\star = \arg\max_{p\notin\mathcal{S}} \left[G(\mathcal{S}) - G\big(\mathcal{S}\cup\{p\}\big)\right], \tag{21}$$

until $|\mathcal{S}| = M$. The map $\mathcal{Q} \mapsto \mathrm{Epis}(\mathcal{Q})$ in (20) is monotone and submodular and the switching term is modular, so (21) attains the standard $(1-1/e)$ guarantee on the information-driven part at $\mathcal{O}(NM)$ score evaluations per slot. The pilot set is then taken as the m activated ports of largest posterior variance,

$$\mathcal{Q}_t = \arg\max_{\mathcal{Q}\subseteq\mathcal{S}_t,\,|\mathcal{Q}|=m} \sum_{n\in\mathcal{Q}}\sum_{k=1}^{K}\left[\boldsymbol{\Sigma}_k\right]_{nn}, \tag{22}$$

which spends the pilot budget where the belief is least certain.

V. Online Doppler Learning

Sections II–IV assume the Doppler rate f_D, and hence $\mathbf{a}$, are known. We now remove that assumption. Since the policy holds ports across slots, the pilots themselves carry lagged pairs from which the temporal autocorrelation can be estimated. For lag τ, let $\mathcal{P}_\tau = \{(t,n) : n \in \mathcal{Q}_t \cap \mathcal{Q}_{t-\tau}\}$ be the co-piloted pairs. The estimator normalizes each lag by the power of the *same* sample pairs,

$$\hat{r}(\tau) = \frac{\sum_{(t,n)\in\mathcal{P}_\tau}\sum_k \Re\{y_{k,n}(t)\,y_{k,n}^*(t-\tau)\}}{\sum_{(t,n)\in\mathcal{P}_\tau}\sum_k \frac{1}{2}\left(|y_{k,n}(t)|^2 + |y_{k,n}(t-\tau)|^2\right) - \sigma_e^2}, \tag{23}$$

where the subtraction removes the pilot-noise floor. Matching the numerator and denominator sample sets is essential: the pooled alternative, in which the denominator averages over *all* samples, is a ratio of sums drawn from different sets, and Jensen's inequality inflates it by $\mathcal{O}(r/n_\tau)$. At the effective sample counts available here (n_τ of order tens) that bias is $+0.034$ in $\hat{r}(1)$, which (23) removes.

Each lag carries an uncertainty, quantified by Bartlett's formula

$$\mathrm{se}(\tau)^2 \;\approx\; \frac{1 + 2\sum_{j<\tau}\hat{r}(j)^2}{n_\tau}, \qquad n_\tau = |\mathcal{P}_\tau|. \tag{24}$$

Feeding $\hat{\mathbf{r}}$ directly into (3) fails: the estimated $\hat{\boldsymbol{\Gamma}}$ is ill-conditioned, and the resulting coefficients are both large in norm and badly overconfident about the innovation variance. We therefore regularize by the estimator's own uncertainty and select the order from the data,

$$\big(\hat{\boldsymbol{\Gamma}}_q + \delta\mathbf{I}_q\big)\mathbf{a}_q = \hat{\mathbf{r}}_q, \qquad \delta = \sum_{\tau=1}^{p}\mathrm{se}(\tau)^2, \tag{25}$$

$$q^\star = \arg\min_{1\le q\le p}\ \mathrm{med}_b\left\{\mathbf{a}_q^{(b)} \text{ incurred error under } \hat{\mathbf{r}}^{(b)}\right\}, \tag{26}$$

where b indexes B bootstrap draws $\hat{\mathbf{r}}^{(b)}$ generated from (24). The selected coefficients are zero-padded to length p. No constant in (25)–(26) is tuned: δ is read off the data through (24), and $q^\star$ is chosen by the incurred error rather than by an information criterion. When the pilots are too few to resolve a high order, (26) returns a low one — refusing the model it cannot support is the mechanism that keeps the closed loop stable.

VI. Numerical Results

[TODO: Setup table; pilot–rate trade-off figure; Doppler-learning figure; baselines.]

VII. Conclusion

[TODO:]

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